SMART INDIA HACKATHON 2026

- **Problem Statement ID –** 26105
- **Problem Statement Title-** AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name -**SecureX

SECURE H

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The Problem :

Cyber risk is still reported as "High/Medium/Low" — qualitative labels that don't translate into financial decisions or budget justifications.

Current Challenges:

- Cannot quantify Return on Security Investment.
- No per-control compliance mapping — ISO 27001, NIST CSF, and CIS Controls exist as PDFs, not live dashboard data
- Risk acceptance lacks auditable evidence.

Constraints:

- DPDP Act (2023): Up to ₹250 Cr per instance of non-compliance.
- RBI & SEBI: Require board-level cyber governance.

Impact:

- Severe tail-risks hidden by subjective scores.
- Weak audit trails invite regulatory failure post-breach.
- Every simulation is unreproducible — no seed, no version, no forensic re-run capability

Proposed Solution

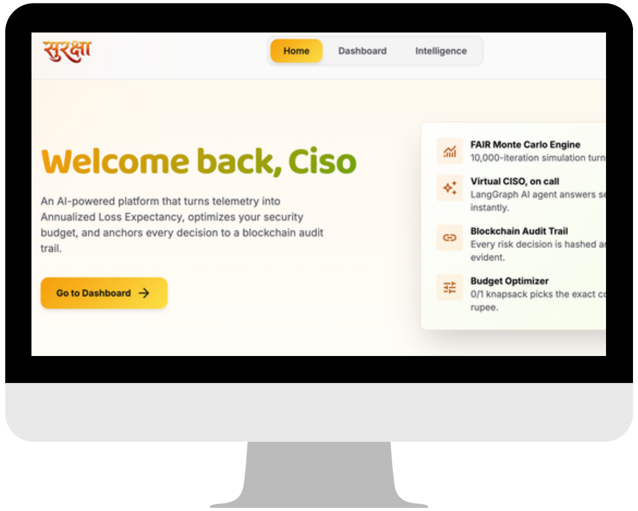
Overview:

Developed an AI-powered Cyber Risk Quantification (CRQ) platform that:

- Automates asset discovery by directly parsing raw network configurations.
- Calculates financial risk (₹ ALE/VaR) using 10k FAIR Monte Carlo simulations and Bayesian self-correction.
- Stores executive risk sign-offs on an Ethereum blockchain for tamper-proof auditing.
- Provides interactive dashboards for loss distribution curves and live SEBI compliance mapping.
- Optimizes security budgets using PuLP Knapsack algorithms to maximize ROSI.

Key Features:

- Financial exposure scoring (FAIR math + Bayesian ML)
- Ethereum hash storage → immutable audit trail, forensic reproducibility
- LangGraph AI Virtual CISO for plain-language reporting
- Training-linked math → employee compliance actively reduces ₹ ALE



Why Different ?

- Replaces subjective "High/Low" scores with exact financial exposure.
- Maps lateral threat blast-radius across the full network topology, not just isolated assets.
- Auto-calibrates against real incidents, permanently eliminating stale assumptions.
- Secures executive risk decisions on a tamper-proof Web3 ledger for regulators.

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TECHNICAL APPROACH

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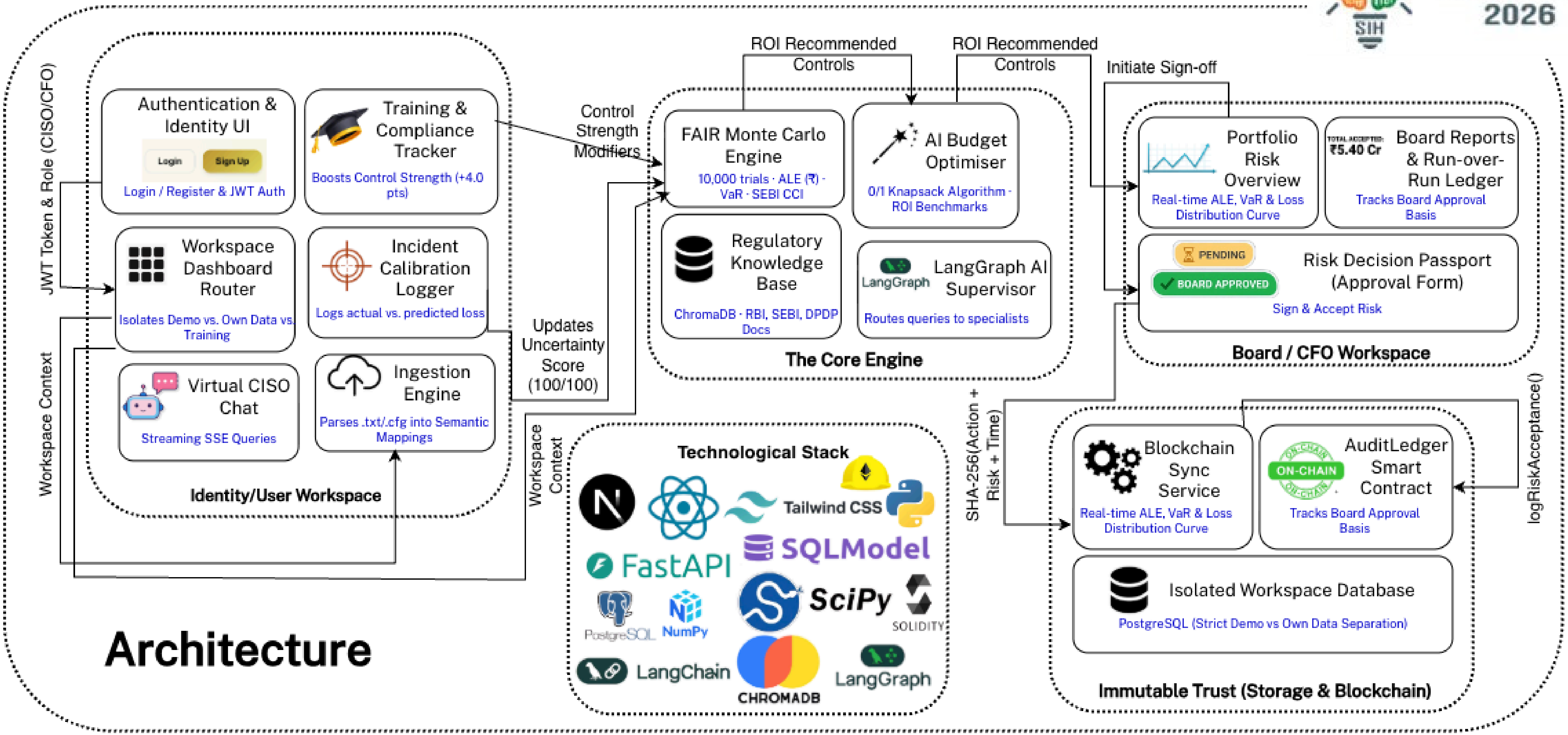

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FEASIBILITY AND VIABILITY

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Operational Feasibility

- Integrates seamlessly with existing enterprise SIEM/CMDBs via a headless FastAPI backend.
- Zero training required for C-suite adoption, thanks to the plain-language Virtual CISO.
- Multi-tenant architecture allows gradual, department-by-department rollout without data leakage.

Integrated Workflow

1. Data Ingestion: >>
Upload raw telemetry or sync with existing asset databases.

2. Risk Modeling: >>
Engine calculates FAIR inputs and blast-radius vulnerability.

3. Quantification: >>
10k Monte Carlo runs generate Rupee-based ALE & VaR.

4. ROSI Optimization: >>
Executives use the "What-If" dashboard to maximize budget efficiency.

5. Web3 Auditing: >>
Approvals and random seeds are hashed to an Ethereum ledger.

6. Auto-Calibration: >>
Real incident outcomes feed back into the Bayesian engine to self-correct.

Technical Feasibility

- 100% open-source data stack (SciPy, NumPy, PuLP) → easy to implement, zero licensing fees.
- Local Hardhat/Ethereum nodes support permissioned, low-cost blockchain auditing.
- Per-control ISO 27001 / CIS Controls v8 / NIST CSF crosswalk maps every sandbox toggle to auditable compliance evidence

Security & Tamper-Proofing

- **Tamper-Proof by Design:** Blockchain stores only deterministic hashes and approval flags — never raw PII — and the smart contract restricts writes to the platform server alone, making external manipulation impossible.
- **Reproducible Math:** Storing the exact simulation seed prevents internal teams from retroactively altering past risk models.

Social & Institutional Viability

- **Aligns with Regulators:** Directly supports SEBI's 5-pillar CSCRF and RBI board-oversight mandates.
- **Bridges the C-Suite Gap:** Translates complex technical metrics into financial language that boards understand.

Economic Viability

Category	Type	Estimated Cost (INR)	Justification / Cost-Cutting
Software Dev	Non-Recurring	₹4Lakhs - ₹5Crore	Open-source stack eliminates enterprise licensing.
Cloud Hosting (AWS)	Recurring	₹1 Lakh / yr	Runs on standard cloud VMs
LLM Inference APIs	Recurring	₹85K/ yr	Usage-based Groq/OpenAI with offline fallbacks.
Web3 Ledger Setup	Non-Recurring	₹50K	Private PoA node or Layer-2 keeps gas fees near zero.
Training Materials	Non-Recurring	₹75K	Built-in 5-module /training page reduces onboarding time.

Total Estimated One Year Cost

Type	Approx. Total
Non-Recurring (One-time)	₹4.5 L –₹5.5 L (~₹5 L avg.)
Recurring (Annual)	₹1.5 L –₹2.0 L (~₹1.8 L avg.)
Total Year 1 MVP Cost	≈ ₹ 6.8 L Total

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IMPACT AND BENEFITS

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Economic & Cost Benefits

- **Cost Savings:** Replaces ₹20+ Lakh annual manual consulting fees with continuous, automated SaaS quantification.
- **Budget Optimization:** Uses a PuLP Knapsack algorithm to mathematically maximize Return on Security Investment (ROSI).
- **Penalty Avoidance:** Proactively maps financial exposure to avoid devastating DPDP Act (2023) whichstates up to ₹250 Cr per instance of non-compliance.

Operational Benefits

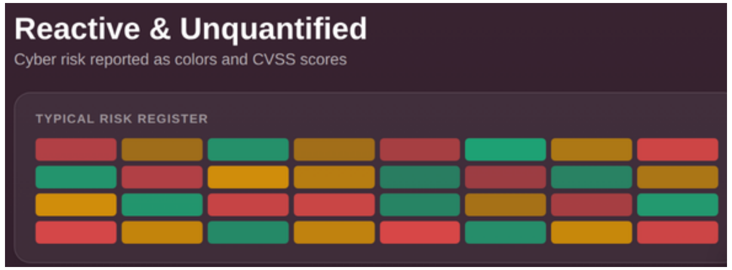
- **Real-Time Agility:** Shifts risk assessments from stale 6-month cycles to daily, real-time simulations.
- **Zero Model Drift:** The Bayesian engine auto-calibrates the risk model using real-world incident data, eliminating static assumptions.
- **Executive Clarity:** The LangGraph AI translates complex technical vulnerabilities into plain, actionable business language.

Social & Regulatory Impacts

- **Data Protection:** Ensures DPDP Act compliance by quantifying PII-linked breach costs before they occur
- **Compliance Readiness:** Natively aligns with RBI/SEBI board oversight mandates and 5-pillar resilience frameworks.
- **Immutable Trust:** The Web3 Ethereum ledger creates a tamper-proof culture of executive accountability.

Future Roadmap

- **Phase 1 (Current):** FAIR Monte Carlo engine, Bayesian calibration, and Ethereum Sepolia testnet auditing.
- **Phase 2 (Integration):** Direct API connectors for Splunk, Microsoft Sentinel, and ServiceNow.
- **Phase 3 (Scale):** Migration of the Audit Ledger to a public enterprise Layer-2 blockchain (e.g., Polygon).
- **Phase 4 (Explainable AI):** Integration of SHAP-based AI for automated mitigation narratives and counterfactual modeling.



BEFORE

AFTER

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RESEARCH AND REFERENCES

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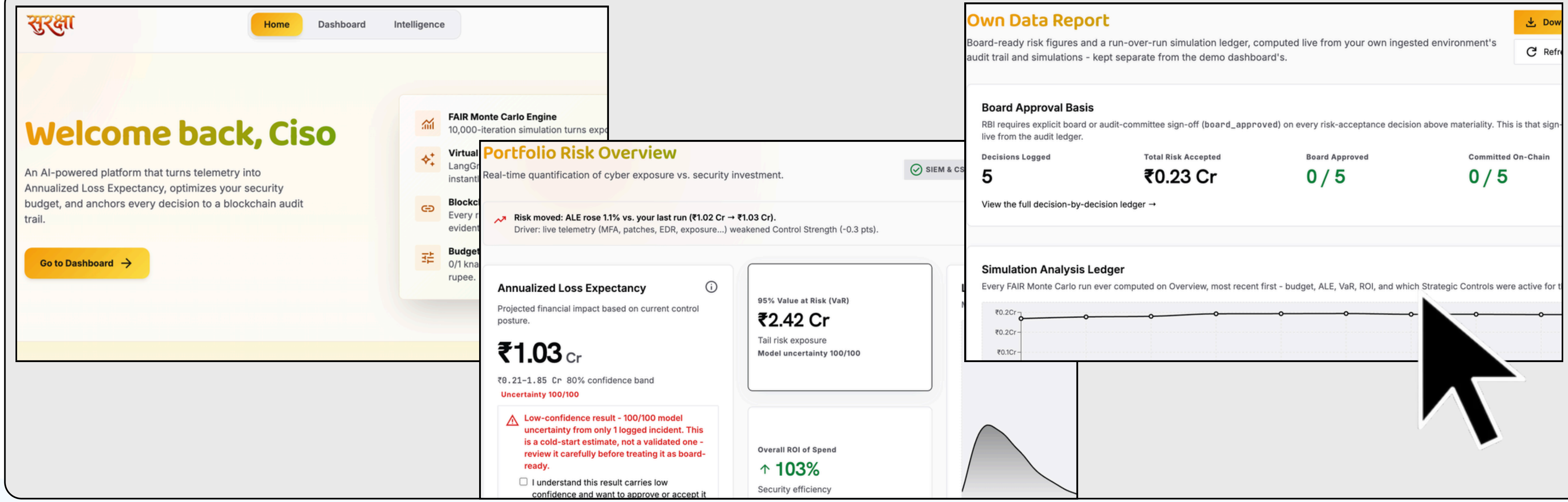
- **PuLP: 011 Knapsack Optimizer** — LINK
- **IEEE Xplore: Bayesian Risk Calibration** — LINK
- **FAIR Model: For Risk Quantification** — LINK
- **Groq + LangGraph: AI Assistant Agent** — LINK
- **SEBI CSCRF: Regulatory Framework** — LINK
- **DPDP Act 2023: Data Protection Law** — LINK
- **NIST & ISO 27001: Security Standards** — LINK
- **Web.py: Blockchain Audit Ledger** — LINK
- **CIS Controls v8: Defensive Safeguards Framework** — LINK



UI/UX